

Functional Principal Component Analysis (FPCA)

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Motivation

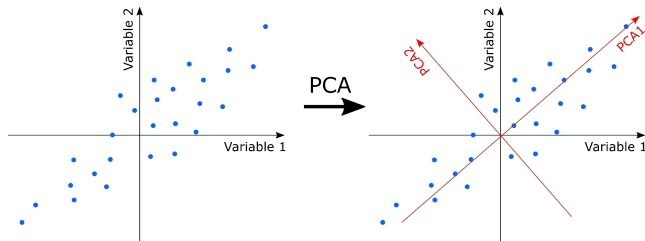
Functional data: each observation is a curve or a function $X_i(t)$.

Examples:

- Growth curves (height vs. age)
- Daily temperature profiles
- Motion trajectories

Goal: extract main modes of variation across functions, analogous to PCA for vectors.

Recap: Classical PCA



- given $x \in \mathbb{R}^d$. If x has covariance matrix Σ , then the variance of projection $u^T x$ is $u^T \Sigma u$.
- Objective: maximize $u^T \Sigma u$ subject to $u^T u = 1$.
- This leads to the eigen-equation

$$\Sigma u = \lambda u.$$

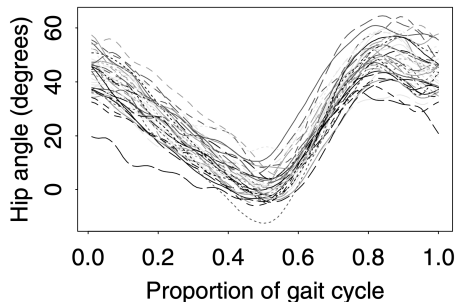
Recap: Classical PCA in practice

- Estimate covariance matrix:

$$\Sigma = \frac{1}{n} \sum_{i=1}^n x_i x_i^T.$$

- Compute the eigen-decomposition: $\Sigma = UDU^T$.
- Columns of U are orthonormal and represent a new basis.
- entries λ_k of D give variances along corresponding directions U .
- The ratio $\lambda_k / \sum_j \lambda_j$ shows the proportion of variance explained.
- The k -th eigenvector u_k is the k -th principal component direction.
- The score for observation x_i on component k is $s_{i,k} = x_i^T u_k$.
i.e. $x_i = \sum s_{i,k} u_k$.

From Vectors to Functions



- Each observation is a function: $X_i(t)$ defined on domain $\mathcal{T}$.
- K FPCs are presented as functions $w_1(t), \dots, w_K(t)$.
- Curves is projected to FPCs as $X_i(t) = \sum_{k=1}^K s_{i,k} w_k(t)$.

Aim: to find principal components and scores.

building FPCA

- Given N data curves $X_i^*(t)$.
- Preprocessing: $X_i(t) = X_i^*(t) - \mu(t)$ with $\mu(t) = \frac{1}{N} \sum_{i=1}^N X_i^*(t)$.
- Find $s_{i,k} := \int w_k(t) X_i(t) dt$ via following problems:

Problem 1:

$$\max_{w_1(\cdot)} \sum_{i=1}^N s_{i,1}^2, \quad \text{s.t.} \quad \int w_1^2(t) dt = 1.$$

Problem k:

$$\max_{w_k(\cdot)} \sum_{i=1}^N s_{i,k}^2, \quad \text{s.t.} \quad \int w_k^2(t) dt = 1,$$
$$\forall j \in [k-1] : \int w_k(t) w_j(t) dt = 0.$$

Functional covariance function $\sigma(s, t) = \text{Cov}(X(s), X(t))$

Its eigen vectors $w_k(\cdot) : \int \sigma(s, t) w_k(t) dt = \lambda_k w_k(s)$.

By Karhunen-Loève decomposition $\sigma(s, t) = \sum_{k=1}^{\infty} \lambda_k w_k(s) w_k(t)$ gives principal components.

Score $s_{i,k} = \int w_k(t) X_i(t) dt$.

$\lambda_k = \text{Var}(s_{i,k})$, represents amount of variation in the direction $w_k(t)$.

Karhunen–Loève Expansion

Under mild conditions:

$$X(t) = \mu(t) + \sum_{k=1}^{\infty} \xi_k w_k(t), \quad \mathbb{E}[\xi_k] = 0, \quad \text{Var}(\xi_k) = \lambda_k.$$

- $w_k(t)$ — orthonormal eigenfunctions of covariance operator.
- ξ_k — uncorrelated functional principal component scores.

This is the infinite-dimensional analog of PCA.

FPCA vs PCA

Property	PCA	FPCA
Data type	vectors $x_i \in \mathbb{R}^p$	functions $X_i(t)$
Covariance	matrix Σ	kernel $\sigma(s, t)$
Eigenproblem	$\Sigma u = \lambda u$	$\int \sigma(s, t) w(t) dt = \lambda w(s)$
Projection	$u^\top x_i$	$\int w(t) X_i(t) dt$
Interpretation	directions in $\mathbb{R}^p$	modes of functional variation

Computing FPCA

Idea: use some basis and express everything through it.

- Given a basis elements $\phi_j(t)$, $j \in J$.
Denote $\phi(t) = (\phi_1(t), \dots, \phi_J(t))^T \in \mathbb{R}^{J \times 1}$ – basis vector at t .
- Denote $x(t) = (x_1(t), \dots, x_N(t))^T \in \mathbb{R}^{N \times 1}$ – vector of observations.
- $x(t) = \mathbf{C}\phi(t)$ – x expressed by basis elements. $\mathbf{C} \in \mathbb{R}^{N \times J}$.
- $\hat{\sigma}(s, t) = \frac{1}{N}x(s)^T x(t) = \frac{1}{N}\phi^T(s)\mathbf{C}^T\mathbf{C}\phi(t)$.
- $w(t) = \mathbf{b}^T\phi(t)$ – vector, expressed via basis.

Eigen problem:

$$\begin{aligned} \max_{\lambda} \int \sigma(s, t)w(t)dt &= \lambda w(s) \\ \text{s.t. } \int w^2(t)dt &= \mathbf{b}^T \left[\int \phi(t)\phi^T(t)dt \right] \mathbf{b} = 1 \end{aligned}$$

Computing FPCA

- $x(t) = \mathbf{C}\phi(t)$
- $\hat{\sigma}(s, t) = \frac{1}{N}\phi^\top(s)\mathbf{C}^\top\mathbf{C}\phi(t)$
- $w(t) = \mathbf{b}^\top\phi(t)$
- $\mathbf{W} = \left[\int \phi^\top(t)\phi(t)dt\right]$

$$\int \sigma(s, t)w(t)dt = \lambda w(s) \Rightarrow$$

$$\int \frac{1}{N}\phi^\top(s)\mathbf{C}^\top\mathbf{C}\phi(t)\phi^\top(t)\mathbf{b}dt = \lambda\phi^\top(s)\mathbf{b} \Rightarrow$$

$$\begin{aligned} \max_{\lambda} \frac{1}{N}\phi^\top(s)\mathbf{C}^\top\mathbf{C}\left[\int \phi^\top(t)\phi(t)dt\right]\mathbf{b} &= \lambda\phi^\top(s)\mathbf{b} \\ \text{s.t. } \mathbf{b}^\top\left[\int \phi^\top(t)\phi(t)dt\right]\mathbf{b} &= 1 \end{aligned}$$

Computing FPCA

- $x(t) = \mathbf{C}\phi(t)$
- $\hat{\sigma}(s, t) = \frac{1}{N}\phi^\top(s)\mathbf{C}^\top\mathbf{C}\phi^\top(t)$
- $w(t) = \mathbf{b}^\top\phi(t)$
- $\mathbf{W} = \left[\int \phi^\top(t)\phi(t)dt\right]$

$$\max_{\lambda} \frac{1}{N}\phi^\top(s)\mathbf{C}^\top\mathbf{C}\mathbf{W}\mathbf{b} = \lambda\phi^\top(s)\mathbf{b} \quad \text{s.t.} \quad \mathbf{b}^\top\mathbf{W}\mathbf{b} = 1$$

Using $\mathbf{u} = \mathbf{W}^{1/2}\mathbf{b}$, and everytime validness for s :

$$\max_{\lambda} \Omega\mathbf{u} = \lambda\mathbf{u} \quad \text{s.t.} \quad \mathbf{u}^\top\mathbf{u} = 1$$

$$\Omega = \frac{1}{N}\mathbf{W}^{1/2}\mathbf{C}^\top\mathbf{C}\mathbf{W}^{1/2}$$

$$\max_{\mathbf{u}} \Omega \mathbf{u} = \lambda \mathbf{u} \quad \text{s.t.} \quad \mathbf{u}^\top \mathbf{u} = 1$$

$$\Omega = \frac{1}{N} \mathbf{W}^{1/2} \mathbf{C}^\top \mathbf{C} \mathbf{W}^{1/2}$$

Let $(\lambda_k, \mathbf{u}_k)$ be an eigenvectors. Then

- $\mathbf{b}_k = \mathbf{W}^{-1/2} \mathbf{u}_k$. $w_k(t) = \mathbf{b}_k^\top \phi(t)$
- $s_{i,k} = \int X_i(t) w_k(t) = \mathbf{b}_k^\top \mathbf{W} \mathbf{C}_i$

To compute parameter C consider, that for each $i \in [N]$ given greed $t_{i,1}, \dots, t_{i,M_i}$. and values $X_i(t_j)$. Then

$$\min_{c_i} \sum_{t=1}^{M_i} \left[X_i(t_j) - c_i^\top \phi(t_j) \right] + \alpha c_i^\top R c_i \Rightarrow \quad (1)$$

$$c_i = [\Phi^\top \Phi + \alpha R]^{-1} \Phi^\top X_i, \quad (2)$$

$$\text{where } \Phi \in \mathbb{R}^{M_i \times J} \quad \text{with } \Phi_{i,j} = \phi_j(t_i), \quad (3)$$

$$X_i = [X_i(t_{i,1}), \dots, X_i(t_{i,M_i})]^\top \in \mathbb{R}^{M_i}. \quad (4)$$

Choice of Basis

Common options:

- **Fourier:** for periodic data.
- **B-splines:** local, flexible for smooth curves.
- **Wavelets:** localized, good for abrupt changes.

B-spline parameters:

- Order (degree p , usually 3).
- Number and placement of knots.
- Optional smoothing penalty $\alpha \int [f''(t)]^2 dt$.

Summary

- FPCA generalizes PCA to functional data in L^2 spaces.
- Computation reduces to a finite-dimensional eigenproblem via basis expansion.
- Choice of basis (Fourier, B-splines, wavelets) controls flexibility.
- Regularization ensures smooth, interpretable eigenfunctions.



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